

THE WEST AFRICAN EXAMINATIONS COUNCIL

BASIC EDUCATION CERTIFICATE EXAMINATION -- JUNE 2008

INTEGRATED SCIENCE -- PAPER 2

Duration: 1 Hour 15 Minutes | Level: JHS 3 / BECE | Code: SCI 01/02 - P2

INSTRUCTIONS TO CANDIDATES

Section A is compulsory and carries 40 marks. Answer any three questions from Section B (carrying 20 marks each).

PART II: ESSAY & STRUCTURED QUESTIONS WITH WAEC RUBRICS

QUESTION 1: Section A: Practical and Experimental Work

In an experiment to test for the presence of starch in a variegated leaf, a student followed standard laboratory procedure.

- (a) State four sequential steps taken by the student to prepare the leaf for testing. [4 marks]

Model Answer / WAEC Rubric: 1. Dip leaf in boiling water for 1-2 minutes to kill cells. 2. Boil leaf in ethanol/alcohol in a water bath to extract chlorophyll. 3. Dip decolourized leaf in warm water to soften it. 4. Spread on a white tile and add drops of iodine solution. [M2, A2]

- (b) Why is the alcohol heated in a water bath instead of directly on a naked Bunsen flame? [2 marks]

Model Answer / WAEC Rubric: Ethanol/alcohol is highly inflammable/flammable and would catch fire on an open flame. [A2]

- (c) State the expected colour change if starch is present in the green portions of the leaf. [2 marks]

Model Answer / WAEC Rubric: Colour changes from yellowish-brown/amber to blue-black. [A2]

- (d) Name three laboratory apparatus required for the simple distillation of saltwater. [3 marks]

Model Answer / WAEC Rubric: Distillation flask, Liebig condenser, Bunsen burner (or thermometer/receiving flask). [B3]

QUESTION 2: Matter and Chemical Reactions

Metals and non-metals display distinct physical and chemical properties in nature.

- (a) Distinguish between an element and a compound, giving one example of each. [4 marks]

Model Answer / WAEC Rubric: An element is a pure substance composed of only one type of atom (e.g. Iron, Oxygen). A compound consists of two or more elements chemically combined in a fixed proportion (e.g. Water, Carbon dioxide). [B2, A2]

- (b) State three physical differences between metals and non-metals. [6 marks]

Model Answer / WAEC Rubric: 1. Metals are good conductors of heat and electricity, non-metals are poor conductors (except graphite). 2. Metals are malleable and ductile; non-metals are brittle. 3. Metals have high melting and boiling points; non-metals have comparatively lower points. [B3, A3]

- (c) Write a balanced word equation for the reaction between dilute hydrochloric acid and zinc granules. [4 marks]

Model Answer / WAEC Rubric: Zinc + Hydrochloric acid → Zinc chloride + Hydrogen gas. [M2, A2]

QUESTION 3: Ecosystems and Soil Management

Soil fertility and conservation are critical for sustainable agriculture in Ghana.

- (a) Explain two ways by which nutrients are lost from agricultural soils. [4 marks]

Model Answer / WAEC Rubric: 1. Leaching: Washing down of soluble mineral nutrients beyond the reach of plant roots by heavy rainfall. 2. Soil erosion: Removal of topsoil containing rich humus by running water or wind. [A2, B2]

- (b) Describe three cultural practices a farmer can adopt to improve soil fertility without synthetic chemicals. [6 marks]

Model Answer / WAEC Rubric: 1. Application of compost or farmyard manure. 2. Crop rotation including leguminous crops to fix atmospheric nitrogen. 3. Mulching to retain moisture, prevent erosion, and add organic matter. [B3, A3]

- (c) State two ecological functions of earthworms in the soil. [4 marks]

Model Answer / WAEC Rubric: 1. Burrowing action aerates the soil and improves drainage. 2. Casts deposited on the surface enrich topsoil with organic nutrients. [A2, B2]